

Q2 - The Midnight Episode: ECG Arrhythmia Detection

'Catching the Arrhythmia' via normalized template correlation (7.5%) - EE200 Course Project

A Holter recording is the discrete-time signal $x[n]$ sampled at $f_s = 250$ Hz with $N = 5000$ samples; in the healthy stretch one full beat {P, QRS, T} arrives every 0.8 s. The strategy: learn one clean beat as a template, slide it along the recording, and flag the moment the shape stops matching. The analysis uses the provided files `patient_ecg.npy` and `template.npy` exclusively; every numeric value below is computed in the notebook, not estimated. Detector functions live in `src/q2_ecg.py`.

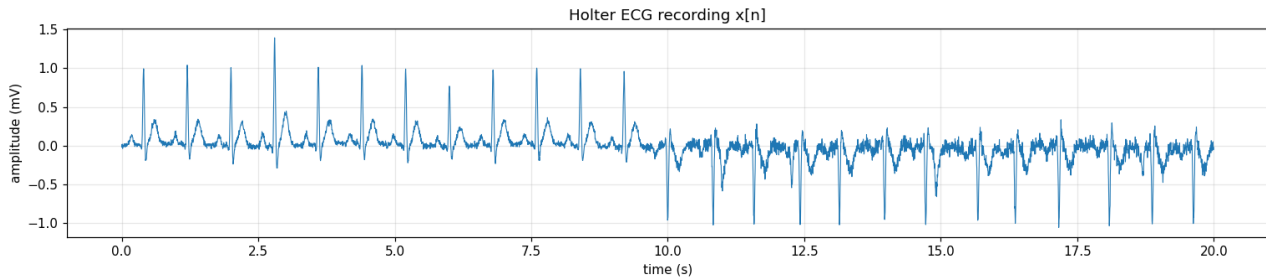


Fig 1. The full 20 s recording: regular, identical beats give way to an irregular region partway through.

(a) Reading the signal [0.75%]

- (i) Duration. The clip length is $N/f_s = 5000/250 = 20.0$ seconds.
- (ii) Heart rate and beat length. One beat every 0.8 s is a rate of $60/0.8 = 75$ beats per minute, and one beat occupies $0.8 \text{ s} * 250 \text{ Hz} = 200$ samples.
- (iii) Fundamental frequency. Treating the healthy ECG as periodic with period $T_0 = 0.8 \text{ s}$, the fundamental frequency is $f_0 = 1/T_0 = 1/0.8 = 1.25 \text{ Hz}$. (Consistently, the 200-sample template is exactly one period long.)

(b) The healthy heart in the frequency domain [0.75%]

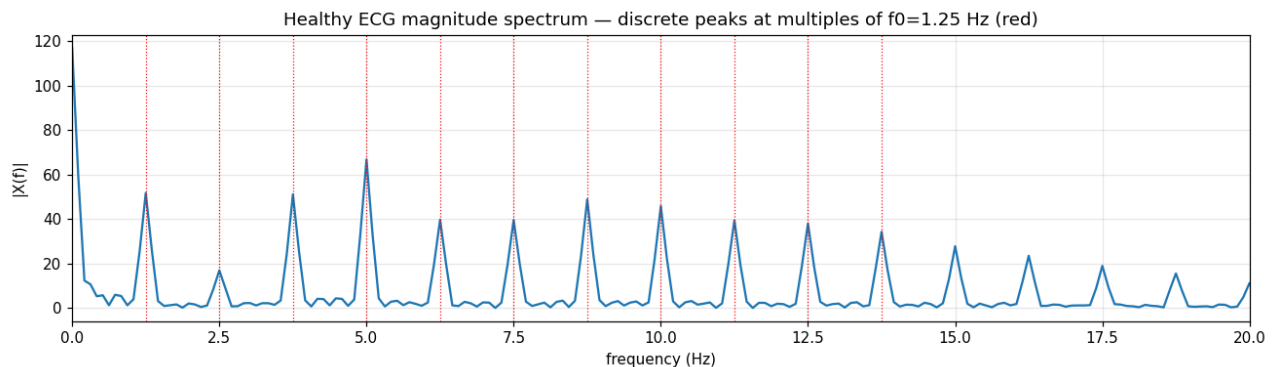


Fig 2. Magnitude spectrum of the healthy segment; red lines mark multiples of $f_0 = 1.25 \text{ Hz}$.

- (i) Shape of $|X(f)|$. A periodic (or nearly periodic) signal does NOT have a smooth continuous spectrum: by Fourier-series theory its energy is concentrated at discrete HARMONICS of the fundamental, f_0 , $2f_0$, $3f_0$, Fig 2 confirms a line/comb spectrum with peaks landing on the red multiples of 1.25 Hz, not a smooth curve.
- (ii) What drives the high frequencies. The QRS complex is a sharp, narrow spike, whereas the P and T waves are broad and smooth. A narrow feature in time requires a broad band of frequencies to synthesise (the time-bandwidth / uncertainty principle), so the QRS is responsible for the high-frequency content; the broad P and T waves are inherently low-frequency.
- (iii) Raising the rate to 150 bpm. The period halves to $T_0 = 0.4 \text{ s}$, so $f_0 = 1/0.4 = 2.5 \text{ Hz}$: the fundamental DOUBLES, and since the harmonic spacing equals f_0 , the spacing between spectral lines doubles as well (the comb stretches out).

(c) Cutting a heartbeat - windowing [1.0%]

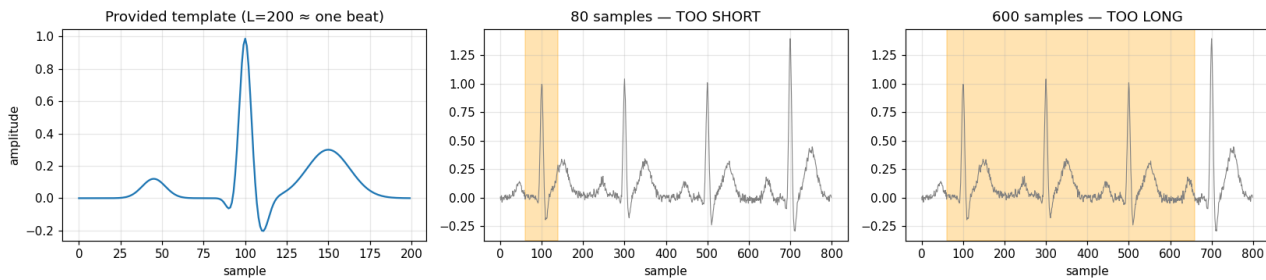


Fig 3. The 200-sample template (one beat) and the two mis-sized rectangular windows on the recording.

Maya isolates a template by multiplying the signal with a rectangular window (1 inside $[n_1, n_2]$, 0 outside).

(i) Correct width and placement. To capture exactly one full beat the window should be about one beat period wide, ~ 200 samples, positioned to bracket a complete $\{P, QRS, T\}$ cycle starting just before the P wave.

(ii) What goes wrong at 80 and 600 samples. An 80-sample window is far too SHORT: it slices into the beat itself, clipping part of the QRS/T so the template is incomplete and distorted, and its hard rectangular edges cut through signal, causing spectral leakage. A 600-sample window is too LONG: at 200 samples/beat it spans roughly three beats, so it swallows fragments of neighbouring beats and contaminates the template with adjacent-beat energy.

(iii) Why shortest is not best. This is the same time-frequency uncertainty trade-off as the STFT: a short window gives sharp time localisation but poor frequency resolution, a long window the reverse. Here making the window 'as short as possible' is actively harmful because it literally cuts through the beat we are trying to model - so the right width is matched to the beat, not minimised.

(d) Matching the template - normalized correlation [1.5%]

$$\rho(m) = [\text{Sum}_k t[k] x[m+k]] / (\|t\| * \|x_m\|)$$

$\rho(m)$ is the cosine similarity between the template t and the length- L segment of x starting at m ; $\|\cdot\|$ is the root-sum-of-squares energy.

(i) Range and perfect-match value. By the Cauchy-Schwarz inequality $\rho(m)$ lies in $[-1, +1]$; $\rho = +1$ signals a near-perfect shape match (template and segment are scalar multiples of each other).

(ii) Why normalize. Real ECG amplitude drifts slowly (baseline wander) and beats vary in size, so we want a score that measures SHAPE, not amplitude. Dividing by the two energies makes ρ invariant to scaling: a perfectly shaped beat that happens to be twice as tall still scores $\rho = +1$. The UNNORMALIZED dot product, by contrast, would roughly DOUBLE for that same beat - it conflates 'right shape' with 'large amplitude', which baseline wander then makes unreliable. (Verified numerically: $\rho(t, 2t) = +1$ while the raw dot product scales by $2x$.)

(iii) An inverted beat. If an abnormal beat is flipped upside-down relative to the template, t and the segment are anti-aligned, so $\rho \sim -1$. Because that is at the opposite extreme from the healthy $\rho \sim +1$, inverted beats are especially easy to flag with a simple threshold.

(e) Onset detection and the spectrogram [1.5%]

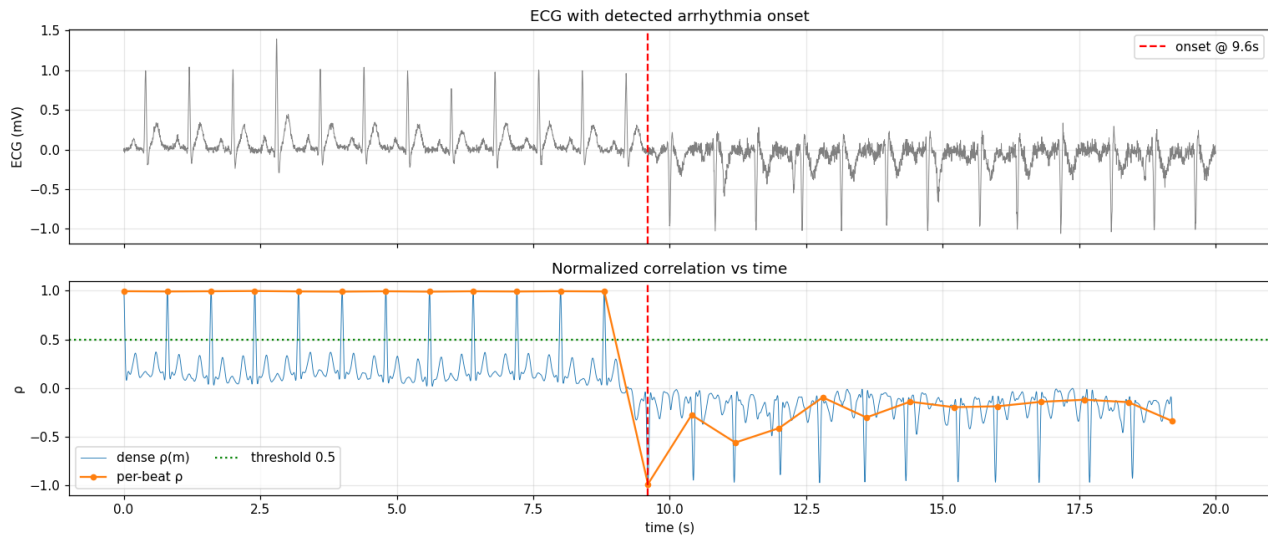


Fig 4. ECG with the detected onset (top) and ρ vs. time (bottom): the first 12 beats score $\sim +1$, then the score drops to ~ -1 .

(i) An onset rule and its threshold trade-off. Declare the arrhythmia onset at the first beat whose correlation falls below a threshold (0.5, as fixed in part g). Setting the threshold too HIGH causes false alarms from normal beat-to-beat variability and noise; setting it too LOW makes the detector under-sensitive and it misses mild but real morphology changes. 0.5 sits safely between the healthy cluster ($\rho \sim +1$) and the abnormal beats.

(ii) The spectrogram view. In the healthy region the spectrogram shows clean, steady horizontal harmonic bands at f_0 , $2f_0$, $3f_0$, ...; in the arrhythmia region those bands smear, break up, shift and are joined by irregular new energy as the periodicity is lost.

(iii) Why they can disagree, and which to trust. The correlation score is evaluated per beat, so it can pinpoint the exact beat where the shape breaks. The spectrogram must accumulate several irregular beats inside its analysis window before the harmonic structure visibly degrades, so its estimate of the onset is inherently smeared and delayed. To pinpoint the single moment a bad beat first appears, trust the CORRELATION; use the spectrogram as the corroborating big-picture view. (This is again the windowing time-resolution trade-off from part (c).)

(f) Sampling and aliasing [0.5%]

The clinically important QRS features contain content up to about 40 Hz.

(i) Nyquist minimum. To capture content up to 40 Hz without aliasing the sampling rate must be at least $2 * 40 = 80$ Hz.

(ii) What 50 Hz does. 50 Hz is below 80 Hz, so any QRS content above the new Nyquist frequency of 25 Hz folds back ('aliases') into lower frequencies, distorting and smearing the sharp spike or injecting spurious low-frequency components. This is dangerous because the detector relies on the SHAPE of the beat through correlation, and aliasing corrupts exactly that shape - causing false positives and false negatives.

(iii) The fix and its cost. If the rate must be lowered, apply an anti-aliasing low-pass filter (cutoff ~ 25 Hz, below the new Nyquist) BEFORE downsampling. The unavoidable cost is that genuine fine QRS detail above 25 Hz is filtered out - real diagnostic morphology is permanently lost in exchange for the lower data rate.

(g) Prototyping the detector: find_onset [1.5%]

`find_onset(ecg_signal, template, threshold)` implements the rule directly: it steps through the recording BEAT BY BEAT (advancing by `len(template) = 200` samples each time, not sample-by-sample), computes the normalized correlation of each beat against the template, and returns the start index of the first beat whose score drops strictly below the threshold; if no beat breaches it, it returns -1. On the provided data

with threshold = 0.5 it returns sample index 2400, i.e. $t = 2400/250 = 9.6$ s (the 13th beat). Read from Fig 4, the first 12 beats score $\rho \approx +0.99$ and the beat at 9.6 s scores $\rho \approx -0.99$ - an INVERTED beat - after which the recording stays irregular. The arrhythmia therefore begins at $t = 9.6$ s. Sanity checks: a threshold above every score returns the first beat, and an impossible threshold returns -1.

(h) Visualising the spectrogram, and choosing nperseg [0.5%]

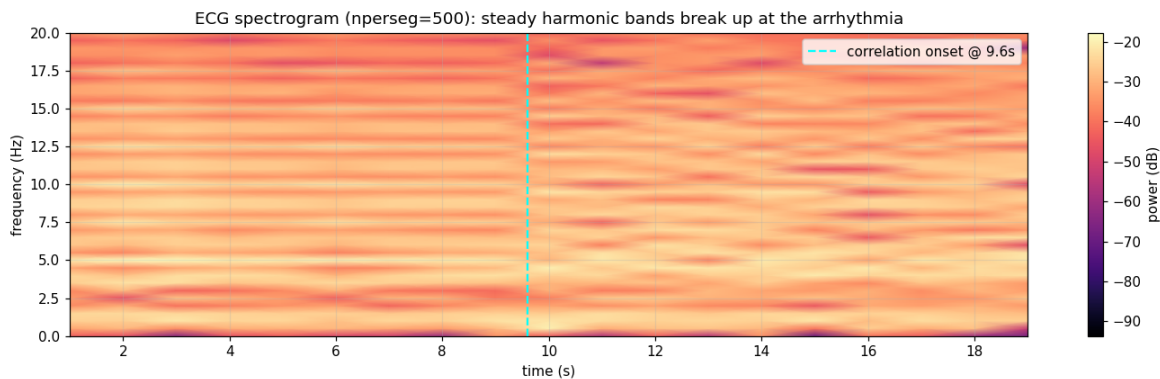


Fig 5. Spectrogram ($nperseg = 500$): steady harmonic bands in the healthy region break up at the arrhythmia onset (marked).

The spectrogram is computed with `scipy.signal.spectrogram` and plotted with `matplotlib`. The critical parameter is the window length `nperseg`, which sets the frequency resolution $df = fs/nperseg$. To resolve the harmonic spacing $f_0 = 1.25$ Hz we need $df < \sim 1.25$ Hz, i.e. $nperseg > \sim fs/f_0 = 250/1.25 = 200$. We choose $nperseg = 500$, giving $df = 250/500 = 0.5$ Hz - comfortably finer than 1.25 Hz - so the harmonics appear as clean, well-separated horizontal bands in the healthy region, exactly as needed to see them break up at the onset. (A much smaller window would blur the harmonics together; a much larger one would smear the onset in time.)

Conclusion (Q2)

On the real recording: 20 s clip, 75 bpm, 200 samples/beat, $f_0 = 1.25$ Hz. The normalized correlation stays at $\sim +1$ for the first twelve beats and then the arrhythmia begins at $t = 9.6$ s (sample 2400) with an inverted beat ($\rho \approx -1$); the spectrogram independently shows the harmonic bands disintegrating at that point. The per-beat correlation pinpoints the exact onset, the spectrogram corroborates it.